

Plasma pressure profiles in a sheared-flow-stabilized Z-pinch

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ABSTRACT

We report the plasma pressure reached inside the central plasma column of a sheared-flow-stabilized Z-pinch using Thomson scattering measurements. Building on previously reported experimental results and the analysis methods established for the high temperature and moderate density plasmas generated on the FuZE device, we show evidence of a central plasma region with higher electron temperature and density, which is consistent with a pinch behavior. Elevated electron temperatures up to 2.25 ± 0.8 keV and densities up to $(4.9 \pm 0.2) \times 10^{17} \text{ cm}^{-3}$ are observed to temporally coincide with the fusion neutron production from the plasma. Reconstructed plasma pressure profiles highlight the presence of a several millimeter-wide column with elevated pressure whose location varies shot-to-shot. The plasma pressure rises as neutron production from the deuterium plasma increases, reaching a peak value of 2.6 kBar. This peak value is consistent with a radially force-balanced pinch equilibrium model based on the measured ~ 320 kA pinch current. Complete datasets were obtained at two axial locations, 10 and 20 cm axial position from the tip of the central electrode, which corroborate the estimated neutron source axial lengths.

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I. INTRODUCTION

Continued advances in inertial and magnetic fusion energy suggest nuclear fusion is a promising candidate for energy production. Hence, various fusion power plant designs are being considered by private industry, academia, or national laboratories across the world. A crucial requirement to understand and improve the performances of fusion experiments is to characterize the plasma conditions reached when fusion reactions occur and produce neutrons. Thomson scattering is one of the gold standard techniques used to measure plasma parameters such as density and temperature and has been deployed on a wide range of experiments from tokamaks to laser facilities.^{1–4} Such a diagnostic has been successfully deployed on the sheared-flow-stabilized Z-pinch,⁵ named FuZE,⁶ which has been operated by the

University of Washington and Zap Energy. Sheared-flow-stabilized Z-pinchs are a candidate for net gain fusion energy production.^{7,8}

Figure 1 shows the temporal evolution of the total current, pinch current, and neutron signal during a typical FuZE discharge using an ignitron-switched capacitor bank charged to 25 kV, with a total stored energy of ~ 70 kJ. The deuterium fill-gas is injected in the chamber by four solenoid injection valves with a plenum pressure set at 200 psi for this discharge. The pinch current is inferred from B-dot probes placed inside the vacuum chamber at many axial and radial positions along the FuZE cylindrical outer electrode. The total current delivered to the device, measured by a Rogowski coil, rises as the plasma is accelerated upstream of the cathode tip down the coaxial axis, similarly to a rail-gun. Because of the shorter central electrode, the current sheath starts

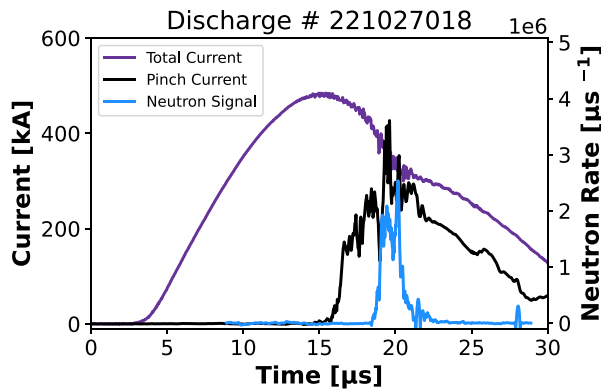


FIG. 1. Temporal evolution of total current, pinch current, and neutron production during a FuZE discharge. The total plasma current in the accelerator (purple) is measured at the upstream anode/cathode gap. The pinch current (black) is inferred from b-dot probes embedded inside the anode. The neutron production rate measured by a plastic scintillator detector is shown in blue.

“zippering” and converging toward the axis. The inductance of the plasma load increases and the total current starts decreasing. At this transition, the pinch current starts rising and the plasma forms a “Z-pinch,” a plasma with axial current confined by its own magnetic field. Detailed description of the gas injection configuration, capacitor bank system, and establishment of the sheared-flow central column can be found in previous publications.^{9,10}

The higher pinch current coincides with fusion neutron generation measured by photomultiplier tubes coupled to scintillator detectors,¹¹ shown in blue. The neutron emission is quantified at five different positions along the pinch axis and several azimuthal positions, which allows an estimation of the axial extent of the neutron producing column,¹² while the total neutron yield is determined by arsenic, yttrium, and bromine activation detectors.

A characterization of the neutron yield alone is not sufficient to help advance the design toward higher fusion efficiency. Neutron yield is a convoluted measurement that nonlinearly depends on many variables, such as density, temperature, and hydrodynamic instability. In this paper, we show how implementing a Thomson scattering diagnostic can help constrain the values of the electron density and temperature and provide insight into the plasma pressure reached in the central column.

II. THOMSON SCATTERING OF THE SHEARED-FLOW PLASMA

The sheared-flow plasmas produced on FuZE are typically on the order of 10–30 cm long, depending on gas puff and bank control parameters, and lasts for several microseconds.⁹ A portable Thomson scattering diagnostic has been deployed on the FuZE device to probe plasma electron density and temperature at different locations. Initial results obtained 20 cm downstream from the central electrode allowed the inference of electron temperature across the plasma radius. It was found that the sheared-flow-stabilized Z-pinch lies in an unconventional regime for Thomson scattering with electron temperatures $T_e \sim 1 - 2$ keV and densities $n_e < 5 \times 10^{16} \text{ cm}^{-3}$. These plasma conditions differ from typical tokamaks, non-collective regime for Thomson scattering, or laser-produced plasmas, collective regime, and therefore

standard analysis for Thomson scattering data is not directly applicable. In Ref. 13, we have described new dedicated analysis tools developed for these sheared-flow-stabilized plasmas. A correlation was established between the observation of elevated electron temperatures and the neutron yield.¹⁴ These results provided only an upper bound on the electron density, leading to less constraints on the form-factor fitting method and significant error bars on the reported temperature. A particular challenge in that study was the variability of the spatial location of the pinch column. This caused movement of the pinch column in and out of the detector’s field of view within a discharge¹⁰ and from discharge to discharge. Hence, many discharges were required to fully sample the plasma column during its compression and confinement stage. To improve on this dataset, the probed location was moved to be 10 cm closer to the tip of the central electrode, this new position is referred to as P10. Here, the variation of the column position is expected to be reduced and density greater. This new probing location, as shown in Fig. 2, is the focus of the work presented in this manuscript.

The Thomson detector relies on an Nd-YAG laser that was frequency doubled to produce 8 J of energy at 532 nm in 1.5 ns. The scattered light from the plasma is collected at 15° from the laser axis and focused by a pair of achromatic doublets onto a linear fiber bundle. The fiber bundle used for this experiment is a linear array of 27 fibers each with a diameter of $100 \mu\text{m}$, center-to-center spacing of $130 \mu\text{m}$, and a numerical aperture of 0.12, or $f/4$. The bundle transports the light to a spectrometer and camera assembly. On the spectrometer side, the fiber bundle has a center-to-center separation of $390 \mu\text{m}$. This larger separation ensures each fiber is individually resolved on the detector. Each fiber collects light from a volume of $0.25 \times 0.25 \times 0.74 \text{ mm}^3$, and the total field of view for all 17 fibers is 16 mm in the radial direction. The spectrometer and the camera are the Andor 500 mm Shamrock Czerny-Turner spectrometer and the Andor iStar CMOS detector, respectively. The 500 mm focal length spectrometer is used with a low resolution grating, 150 lines/mm, to observe the electron feature of the Thomson scattered spectrum. The camera exposure length is 4 ns to capture the entire pulse width while minimizing the background light emission coming from the plasma and the sensitivity

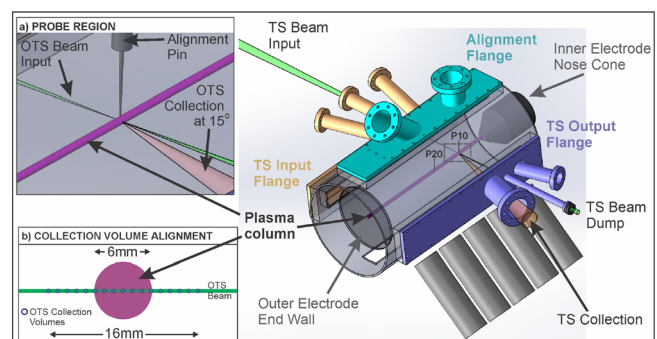


FIG. 2. Experimental setup of the Thomson scattering diagnostic on the FuZE device. The probe laser, labeled TS Input beam, is entering the chamber perpendicularly to the plasma column to probe plasma conditions 10 cm away from the central electrode. The plastic scintillator detectors are shown as the gray cylinder under the collection flange for Thomson scattered light. (a) Inset is a zoomed-in view of the probe laser, plasma column, and collection angle. (b) Inset shows the 17 volumes probed in the radial direction across the column.

to fluctuations in the timing between the laser and the camera exposure.

III. RECONSTRUCTING THE ELECTRON TEMPERATURE AND DENSITY

Thomson data from 104 discharges at the new P10 probe location are presented in this work. Figure 3 shows an example of raw spectra for each of the 17 fibers for a single discharge. The signal at the laser wavelength is rejected by a notch filter to attenuate parasitic light coming from various reflections of the probe laser inside the vacuum vessel that can be collected by the detector. The filter rejection band is centered near the laser wavelength, and it has a full width at half maximum of 17 nm. The angle of incidence on the notch filter was set to 18° to shift the center of the rejection band to 528 nm and maximize the red-shifted signal while still obscuring the parasitic noise at the central laser wavelength. The raw Thomson spectrum width is broad enough to be observed on both sides of the notch filter. This is a major difference compared to previous results obtained 20 cm away from the tip of the central electrode where only the red-shifted peak was consistently above the background. This broader signal can be explained by a larger electron density that increases the signal level as well as an increased temperature, which we will detail in the following form-factor fit discussion.

Another notable difference in the spectral data between this dataset at P10 and that of P20 is the correlation between the brightness of the signal and its width. Figure 4 shows the evolution of the Thomson scattered spectra at five distinct radial locations by summing the signal over 40 lines of pixels. The Thomson spectrum collected at $r = -5$ mm (blue trace) shows the smallest peak signal and width; the blue-shifted contribution (500–520 nm) is barely above the background level. Conversely, signal collected from $r = 0$ mm is brighter and wider, showing measurable signal above the background on the blue-shifted side. At P20, we had consistently observed that the signal level would decrease with higher inferred temperature, i.e., broader spectra. The trend observed at P10 appears consistent with a pinch behavior—electron density is increasing with temperature—and suggests we may be able to infer both the density and temperature.

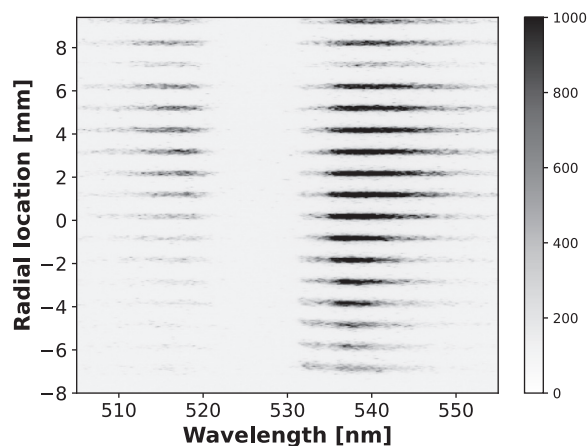


FIG. 3. Raw Thomson scattered spectra measured 10 cm away from the central electrode at the start time of the neutron pulse, t_s (30% of peak emission) for discharge 221 026 059.

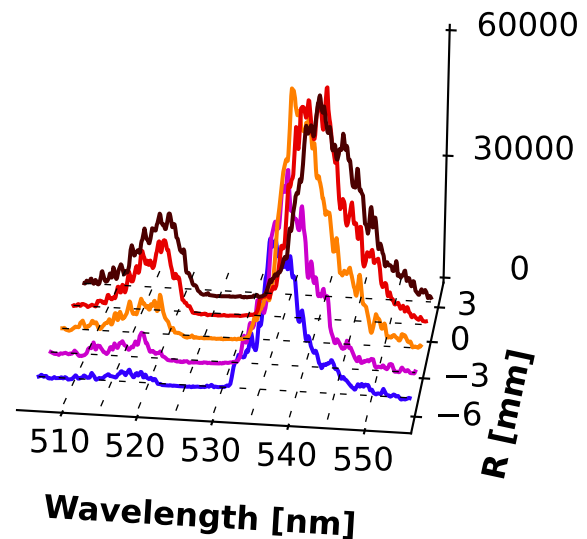


FIG. 4. Experimental spectra summed over each fiber signal's vertical extent (shown in Fig. 3) at five different fiber locations.

Following the method detailed by Banasek *et al.*, we fitted the experimental data using a non-collisional Thomson scattering model. The model has four parameters: T_e , n_e , the signal peak intensity, and continuum intensity. The continuum intensity is a vertical offset of the profile to account for background emission of the plasma. Though velocity can create a Doppler shift in the collected spectrum, the maximum velocity of 200 km/s measured on previous experiments¹⁰ would result in only a shift of $0.9 \text{ \AA}$ due to our choice of scattering angle. Therefore, velocity is not used as a free parameter in the analysis.

Figure 5 shows the fit using either the red-shifted side only (535–570 nm) or both sides of the scattered spectrum (500–570 nm) and how they compare to the experimental data. The experimental error bar, σ , is calculated using $\sigma^2 = yG + n\sigma_{pix}^2$ with y the number of counts for each wavelength, $G = 1000$, the gain of the camera, n , the number of rows used to create the line-out, and $\sigma_{pix} = 11$, the dark noise measured on the camera. There is good agreement between the form-factor fit and the experimental data, as the fit for both cases lies within the error bars. Using one or both sides of the spectrum, the inferred electron density value is the same. The electron temperatures differ from 1.7 keV when using only one side to 1.8 keV when using both sides.

To go further, Fig. 6 compares the reported values for density and temperature for each fiber location using either both sides or only the red-shifted side of the spectrum. Similarly to the previous plot, both methods agree within error bars despite the order of magnitude variation in n_e across the field of view. The same analysis was conducted for a dozen shots, showing similar agreement. Inferred values presented in the rest of the paper will rely only on the red-shifted peak to improve consistency across the two probing locations and the different probe times. This choice was made because at late probe times, $t \geq 1.5 \mu\text{s}$, an emission line at 528 nm starts to overtake the background signal and complicates the Thomson spectrum analysis. This signal has been quantified by collecting data on shots where the probe laser was turned off. It confirmed the signal from this emission line is not broad enough

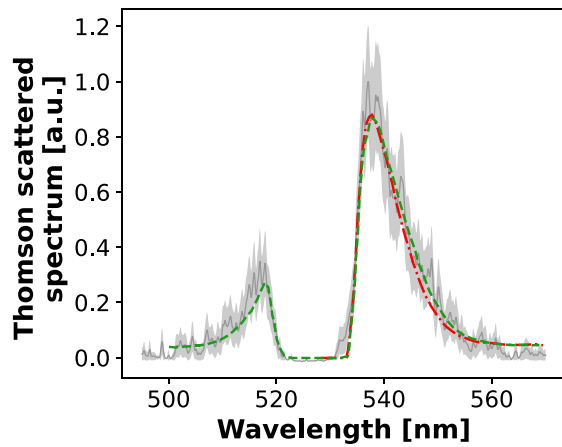


FIG. 5. Best form-factor fit for the fiber at $r=5$ mm on discharge 221 027 063. Experimentally measured Thomson scattered spectrum is represented in gray with error bars symbolized by the gray-shaded area. The convolution of the form-factor data with the notch filter transmission are represented by the dashed red and green dash-dotted lines. The green fit uses both sides of the spectrum and the red one uses only the red-shifted side. The inferred $n_e = 1 \times 10^{17} \text{ cm}^{-3}$, $T_e = 1.8 \text{ keV}$ when both sides of the spectrum are used and $n_e = 1 \times 10^{17} \text{ cm}^{-3}$, $T_e = 1.7 \text{ keV}$ with only the red-shifted part of the spectrum.

to be observed on the red-shifted peak on the other side of the notch filter.

Figure 7 regroups all the inferred n_e , T_e throughout the temporal envelope of the neutron pulse for three different plenum pressures of 150, 200, and 250 psi used for the gas injection valves. In each case, the time probed by the Thomson scattering laser is quoted with respect to

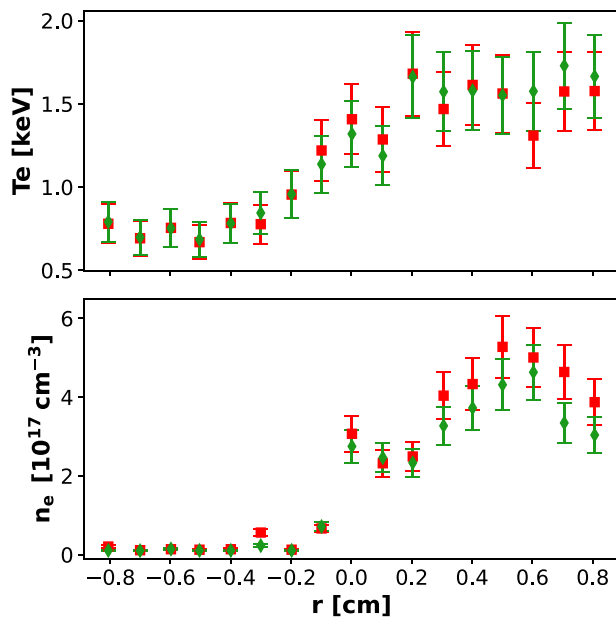


FIG. 6. Inferred value of electron temperature (top) and density (bottom) using only the red-shifted part of the spectrum (red squares) or both red- and blue-shifted parts of the spectrum (green diamonds) for discharge 221 026 059 presented in Fig. 3.

the start time of the neutron generation, t_s , defined as 30% of the peak neutron signal. For this study, t_s is more relevant as a timing fiducial because neutrons should only be produced when the pinch is formed. On the other hand, the peak value may be reached at different times depending on when the pinch column breaks apart due to hydrodynamic instabilities growth, which is less reproducible temporally. The delay between the plastic scintillator detectors and the laser has been characterized to within 5 ns, by firing the probe laser inside the PMT tubes, and also accounts for the jitter in the capacitor bank discharge of $\sim 1 \mu\text{s}$. The electron temperature rises above 1 keV as soon as the neutron start time is reached for the three different cases. Elevated temperature lasts for $\sim 1.5 \mu\text{s}$. The lower gas pressure case shows significant variations of the temperature values throughout the neutron pulse with peak temperature of $2.25 \pm 0.8 \text{ keV}$. Simultaneously, the density peaks at $(4.3 \pm 0.8) \times 10^{17} \text{ cm}^{-3}$ between 0.4 and 1.2 μs . Results at 200 psi, center panel of Fig. 7, also show elevated temperature during the neutron production with a qualitative trend: the temperature rises up to $2.4 \pm 0.3 \text{ keV}$ close to the t_s and then a continuous drop is observed. The density peaks at $(4.9 \pm 0.2) \times 10^{17} \text{ cm}^{-3}$ close $t = 0$ and subsequent elevated density are observed for $\leq 1 \mu\text{s}$. At the highest gas pressure, the temperature shows no clear trend with a peak value at $1.5 \pm 0.1 \text{ keV}$. The density does not seem to fluctuate outside the error bar throughout the neutron pulse. This broad scan suggests optimal mass injection lies within the gas valve plenum pressure range between 150 and 200 psi. Above this gas pressure, pinch compression appears sub-optimal as the maximum temperature reached is lower and the density does not increase during neutron production.

The shot-to-shot variation of the pinch position with respect to our probe location, albeit reduced compared to results at P20, is still significant. The low sensitivity of our measurements to density fluctuations around $n_e \sim 10^{17} \text{ cm}^{-3}$ limits any further analysis of the confinement quality. Therefore, we focus on discharges with the clearest evidence that the pinch column was captured within the detector's field of view. We down-select the discharges using the following criterion on the data: the temperature and density have to show a spatial structure across the field of view with a clear peak as seen in Fig. 4. This implies the column is getting denser and hotter, which is expected only when the diagnostic probes inside the pinch column. We chose this method because the real estate taken by the Thomson setup and plastic scintillator detectors made it impossible to measure the column location with another optical diagnostic such as interferometry on these shots. In the remainder of this paper, we discuss the ten discharges where the peak as well as the sum of the Thomson scattered signal is not decreasing for increased spectral width.

IV. PLASMA PRESSURE PROFILES

Establishing plasma pressure profiles⁹ of the pinch can help quantify the confinement generated by the sheared-flow-stabilized pinch. Such profiles can be used as a starting point for reduced-order multi-fluid plasma hydrodynamic models to describe the basic physics of the pinch formation processes and bulk characteristics of the plasma. These models have proven very useful in their ability to give a first-principles understanding of the pinch formation process and radially averaged plasma quantities in terms of characteristic device parameters.^{16–18} Assuming the system is only made of deuterium, the electron density and temperature reported in Sec. III can be used to estimate the plasma pressure, $P_{\text{plasma}} = 2nkT$, reached inside the central column of the pinch, with the quasi-neutral plasma density

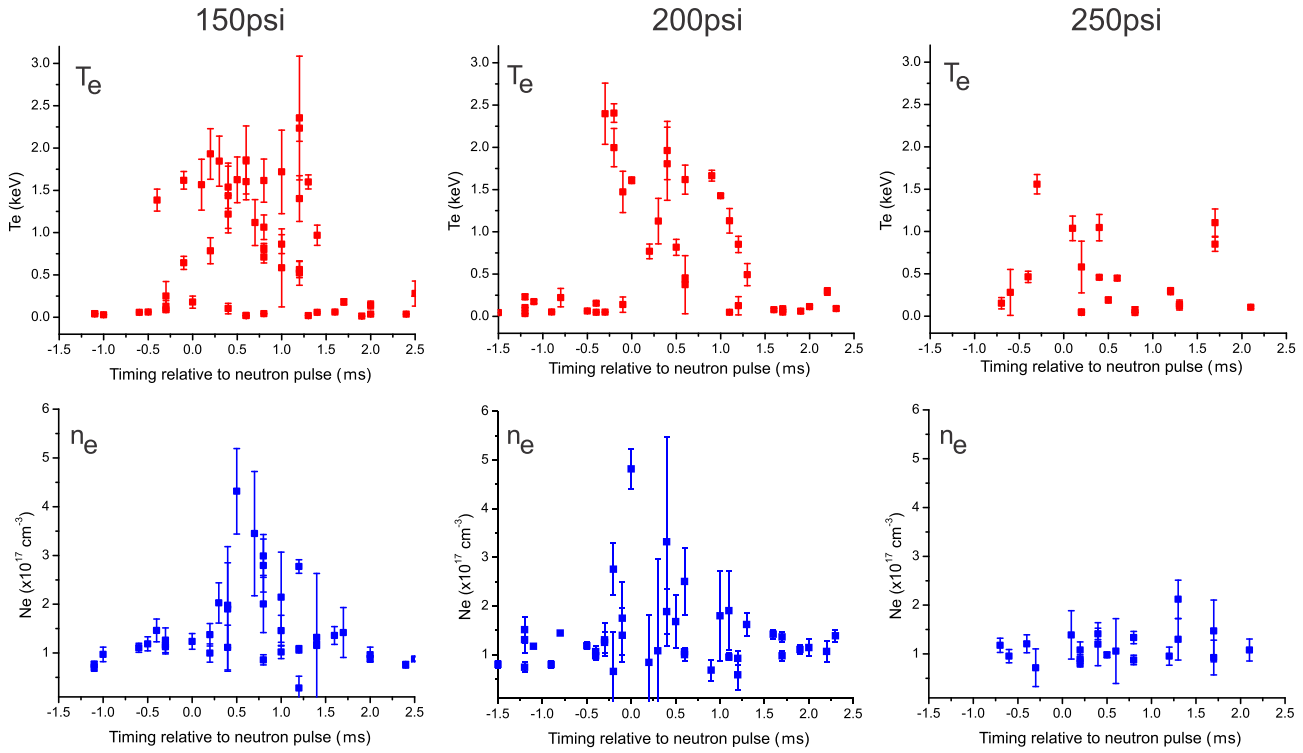


FIG. 7. Top row shows every inferred electron temperature for the three different plenum pressures used during this campaign and bottom row consists of all the corresponding electron densities. Each point corresponds to the average over the four highest temperatures across the 17 fibers. Data are ordered with respect to the timing of the Thomson scattering probe compared to t_s of the neutron emission, defined as 30% of the peak signal.

$n = n_i \sim n_e$, the Boltzmann constant k , and the plasma temperature $T = (T_e + T_i)/2$.

In the FuZE experiment, the electron and D^+ ion equilibration time¹⁹ can be estimated by solving

$$\frac{dT_\alpha}{dt} = \sum_{\beta} \nu_e^{\alpha\beta} (T_\beta - T_\alpha),$$

where $\nu_e^{\alpha\beta} = 1.8 \times 10^{-19} \frac{(m_\alpha m_\beta)^{1/2} Z_\alpha^2 Z_\beta^2 n_\beta \lambda_{\alpha\beta}}{(m_\alpha T_\beta + m_\beta T_\alpha)^{3/2}} \text{ (s}^{-1}\text{)}$, m_β , Z_β , $n_\beta = 5 \times 10^{17} \text{ cm}^{-3}$ correspond to the electron mass, charge, and density, respectively, m_α , Z_α correspond to the ion mass and charge, respectively, and $\lambda_{\alpha\beta}$ is the Coulomb logarithm. Using initial conditions $T_e = 2T_i/3 = 1 \text{ keV}$ and $T_e = T_i/4 = 0.5 \text{ keV}$, it takes, respectively, 2.8 and 4.3 μs to reach $T_e > 0.9T_i$. These equilibration times are comparable to the pinch duration, hence the assumption $T_e \sim T_i$. It is beyond the scope of this work to further investigate temporal equilibration between ion and electron temperature but independent measurements of ion temperature over microsecond long integration time using ion Doppler spectroscopy of carbon^{14,20} showed $T_i \sim T_e$ and will be the subject of a future publication. In addition, the sheared-flow-stabilized Z-pinch formation behavior is expected to adiabatically compress the plasma, which should lead to comparable heating of the ion and electrons.⁶ In the following, the validity of this assumption is tested by comparing the pinch current and total neutron yield using n_e , T_e , and T_i from these measurements to

independent characterization of yield and pinch current using activation detectors and B-dot probes.

Figure 8 shows the plasma pressure profile for a few of the down selected discharges using the previously established criterion. On each profile, we observe the presence of an elevated pressure region smaller

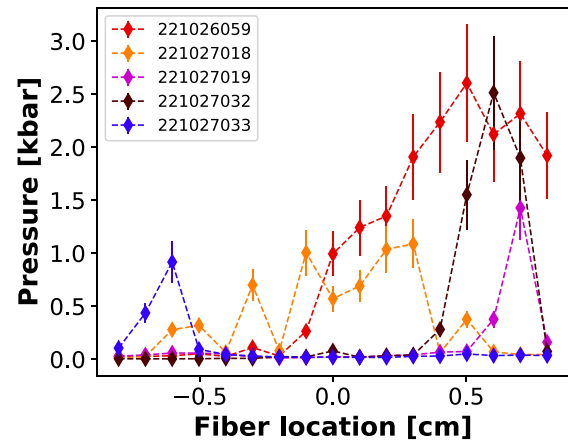


FIG. 8. Inferred pressure along the TS diagnostic line of sight using the electron density and temperature pairs from the form-factor fit for five discharges showing the changing position of the elevated pressure across the field of view.

than the detector's field of view. The diameter and center location of that region vary from a few millimeters to ~ 1 cm. These results corroborate the presence of a central column inside the FuZE device where the plasma is confined. The column location does vary discharge to discharge. In addition to showing the size of the column and the peak pressure, it is instructive to look at the evolution of the peak pressure compared to t_s .

Figure 9 plots the peak pressure compared to the start time of the neutron emission, t_s . Using the reduced dataset, the variation of the peak pressure as a function of time for the three plenum pressures can be investigated. In the 200 psi case, the plasma pressure is continuously increasing and reaches its maximum value of ~ 2.5 kbar at $\sim 0.5 \mu\text{s}$. A similar trend is observed at 150 psi, albeit with limited number of points, while the 250 psi data do not show an increase in pressure. These results show the great potential of measuring plasma conditions inside the central column of FuZE. The peak pressure estimated with TS can be compared to other experimental measurements such as pinch current and neutron production time history to produce a more complete picture of the plasma behavior. TS data collected for delays greater than $0.5 \mu\text{s}$ did not meet the criteria for reconstructing pressure because inferred densities decreased close to the lower limit of our resolution.

V. INFERRING PLASMA COLUMN DYNAMICS

The temporal evolution of the pressure at the P10 location can be compared to the neutron emission measured from the FuZE device. The averaged neutron waveform shape based on the plastic scintillator data taken at $z = 5.33$ cm for 27 discharges using a 200 psi plenum pressure is shown in Fig. 9. The rise in neutron emission coincides with the increase of observed plasma pressure. The signal temporal width is similar to the elevated temperature in Fig. 7,¹⁴ providing additional evidence that the plasma conditions inferred by the TS diagnostic are correlated with the fusion yields and could provide insight into the underlying physics of the FuZE plasmas.

Further validation of the reported pressure can be established by comparing the magnetic pressure expected from the pinch current with the plasma pressure from Thomson scattering. Starting with a

pinch equilibrium,²¹ the magnetic pressure at stagnation follows $P_B = \mu_0 I_{pinch}^2 / (8\pi^2 r^2)$ where I_{pinch} is the current flowing through the pinch and r is the column diameter at stagnation.²² Then, assuming the plasma pressure equals the magnetic pressure, the equivalent pinch current is $I_{pinch} = \sqrt{P_{plasma} \times (8\pi^2 r^2) / \mu_0}$. Using the peak plasma pressure reported of 2.6 ± 0.6 kbar, we find $I_{pinch} = 385 \pm 44$ kA. Such a pinch current is comparable to the one inferred using b-dot probes measurements, on all the 25 kV–200 psi discharges, of 320 ± 80 kA (black curve in Fig. 1 shows one discharge) and provides further validation that the Thomson inferred conditions agree with expectations based on other experimental measurements.

Using the same assumption that the electron and ion temperature are similar, it is possible to estimate the total neutron yield,^{9,23}

$$Y_n = \frac{l\tau}{2} \int n_D^2 \langle \sigma v \rangle_{DDn} 2\pi r dr,$$

with l the spatial extent of the neutron production column, τ the duration of the neutron production, n_D the ion density, $\langle \sigma v \rangle_{DDn}$ the reactivity of the branch of the DD fusion reaction releasing a neutron and r the radius of the plasma column. The density profile assumes a Bennett equilibrium $n(r) = n_0 / (1 + r^2/r_0^2)^2$ with the peak density set at $5 \times 10^{17} \text{ cm}^{-3}$ and a radius $r_0 = 4$ mm. The neutron source characteristics were obtained using the scintillator detectors: $l = 15$ cm and $\tau = 2 \mu\text{s}$ (cf. next paragraph). The predicted total neutron yield is found to increase from 5×10^7 neutrons to 5×10^8 for temperatures between 1 and 2 keV, which is in good agreement with the experimental measurements. Indeed, LaBr activation detectors with yttrium caps reported an average total neutron yields $(5.0 \pm 1.6) \times 10^7$ over all the 25 kV discharges using a 200 psi plenum pressure, and the uncertainty is the standard deviation from the mean.

The observations from the TS diagnostic at p10 and p20 show some interesting differences that may be explained by the spatial extent of the neutron production column. An array of plastic scintillator detectors placed around the experiment is used to estimate the longitudinal extent of the neutron source. Figure 10 shows the results for 150

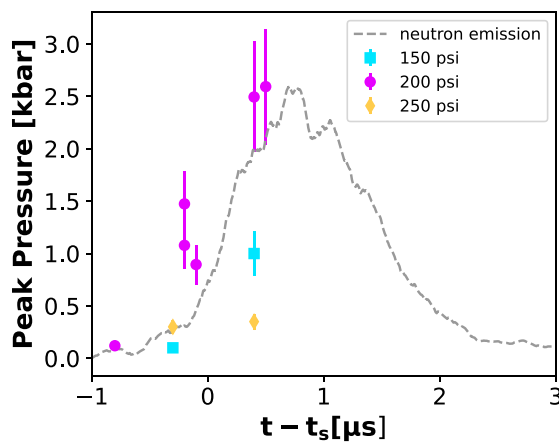


FIG. 9. Temporal evolution of the peak plasma pressure with respect to the neutron signal start time, t_s . An averaged neutron pulse waveform using all the 200 psi discharges is plotted in dashed gray.

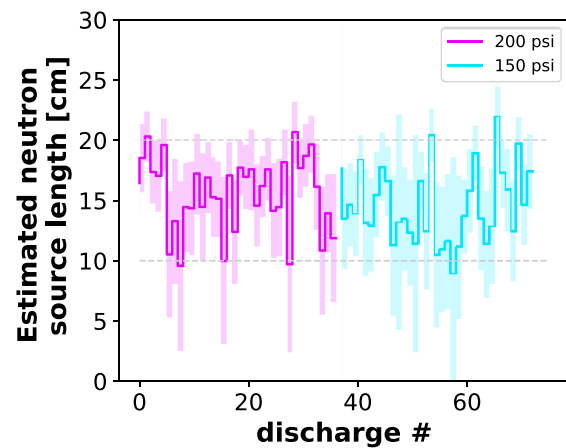


FIG. 10. Axial extent of the line neutron source size reconstructed using the scintillator data for 150 (cyan) and 200 psi (magenta) plenum pressures. Both locations probed by the Thomson scattering diagnostic at P20 and P10 are represented by the two dashed gray lines.

and 200 psi discharges obtained using the analysis method described by Mitrani *et al.*¹² It is worth noting the source length is consistently shorter than 20 cm for the set of machine parameters used in this campaign. Both probing locations for the Thomson scattering diagnostics are shown as dashed gray lines in the figure to illustrate the increased possibility of probing the neutron generating part of the plasma at P10 compared to P20 in both 150 and 200 psi cases. It is beyond the scope of this publication to further investigate the variations of the neutron source length. This dataset prevents stronger conclusions on the temporal evolution of the plasma conditions after peak neutron emission due to the low “catch rate” of the central plasma column within the current field of view.¹⁴ There are many upgrade possibilities to remove these limitations. The Thomson scattering diagnostic is now part of the core suite of diagnostics on FuZE and a duplicate of the current diagnostic is being built for the higher performance FuZE-Q device.⁸ Additional collection angles, a larger field of view and measurements of T_i are also being considered.

VI. CONCLUSION

In this paper, we demonstrated how Thomson scattering measurements can help characterize and guide the sheared-flow-stabilized Z-pinch on FuZE toward improved fusion performances. Moving the probed location closer to the central electrode showed increased density and helped reduce the shot-to-shot variation observed in the previously published data. In addition, the electron density is increasing with temperature, which is consistent with a pinch behavior. Elevated electron temperatures up to 2.25 ± 0.8 keV and densities up to $(4.9 \pm 0.2) \times 10^{17} \text{ cm}^{-3}$ are observed to temporally overlap with the neutron production from the FuZE plasma. This initial characterization of plasma conditions inside the central plasma column is key to compare to existing hydrodynamic multi-fluid plasma model and to guide the design toward a more favorable parameter range. For instance, it allowed for down selecting the most promising plenum pressure required for a 25 kV discharge. Further characterization of the pinch is possible by reconstructing the plasma pressure radial profile seen by the detector. The profiles highlighted the presence of a column with elevated pressure, rising with neutron production and reaching peak conditions consistent with the pinch current reported by b-dot probes. The difference in the two axial locations probed with Thomson scattering is also consistent with the estimated neutron source length reported by plastic scintillator detectors. These results provide unprecedented insight into the FuZE device behavior and demonstrate great potential. The current limitation is the limited amount of data caused by variation in the location of the central column. This is being addressed with added lines of sight and will help provide further understanding of the pinch behavior along the z axis.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

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DATA AVAILABILITY

The data that support the findings of this study are available from Zap energy Inc. Restrictions apply to the availability of these data, which were used under license for this study. Data are available from the authors upon reasonable request and with the permission of Zap Energy.

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